AHMET KAAN AYDIN

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Google Scholar | LinkedIn | ResearchGate | GitHub

EDUCATION

Ph.D. in Applied Mathematics | University of Maryland, Baltimore County | 2022 – May 2027 (expected)
- Advisor: Bedřich Sousedík
- **Awards:** Outstanding Graduate Teaching Assistant

M.Sc. in Applied Mathematics | University of Maryland, Baltimore County | 2024
- Earned en route to Ph.D.

M.Sc. in Mathematics | Western Kentucky University | Bowling Green, KY | 2020 – 2022
- Thesis "Robust Sensor Design for the Novel Reduced Models of the Mead-Marcus Sandwich Beam Equation" (Advisor: Ahmet Özkan Özer)
- **Awards:** John D. Minton Award | Outstanding Graduate Student Award

Pedagogical Formation | Istanbul University | Istanbul, Turkey | 2018 – 2019

B.Sc. in Mathematics | Yildiz Technical University | Istanbul, Turkey | 2015 – 2019
- Senior Project "Affine, and Projective planes"

PUBLICATIONS

- **A.K. Aydin,** A.Ö. Özer, J. Walterman, Robust Model Reductions for the Boundary Feedback Stabilization of Magnetizable Piezoelectric Beams, **Evolution Equations and Control Theory,** vol. 21, pp. 244-276**,** 2026.
- A.Ö. Özer, **A.K. Aydin,** R. Emran, Exponential Stability and Design of Sensor Feedback Amplifiers for Fast Stabilization of Magnetizable Piezoelectric Beam Equations, **IEEE Transactions on Automatic Control,** vol. 70, pp. 1374-1379, 2025.
- **A.K. Aydin**, Md Z. Haider, A.Ö. Özer, A New Semi-Discretization of the Fully Clamped Euler-Bernoulli Beam Preserving Boundary Observability Uniformly, **IEEE Control Systems Letters**, vol. 8, pp. 2955-2960, 2024.
- A.Ö. Özer, **A.K. Aydin**, A novel sensor design for a cantilevered Mead-Marcus-type sandwich beam model by the order-reduction technique, **ESAIM: Control, Optimisation and Calculus of Variations**, vol. 29, article 74, 2023.
- **A.K. Aydin**, A.Ö. Özer, J. Walterman, A Novel Finite Difference-Based Model Reduction and a Sensor Design for a Multilayer Smart Beam With Arbitrary Number of Layers, **IEEE Control Systems Letters**, vol. 7, pp. 1548-1553, 2023.
- A.Ö. Özer, **A.K. Aydin**, Robust-Filtering of Sensor Data for the Finite Difference Model Reduction of a Piezoelectric Sandwich Beam, **61st IEEE Conference on Decision and Control (CDC)**, Cancun, Mexico, 2022.

Submitted and Under Revision

- **A.K. Aydin**, B. Sousedík, A Mean-Informed Low-Rank Monolithic Stochastic Galerkin Solver for the Unsteady Navier-Stokes Equations, arXiv:2608.07526, 2026. Under review at **SIAM Journal on Scientific Computing**.
- N. Akhavan, A.S. George, **A.K. Aydin**, M. Starz-Gaiano, B.E. Peercy, Cell Migration Boundary Motion in Drosophila Egg Chambers: A Combined Phase Field and Chemoattractant Model, under revision.

In Preparation

- **A.K. Aydin**, J. Li, Neural Reduced-Order Approximation of Parametrized Stochastic Galerkin Solutions, in preparation.
- **A.K. Aydin**, B. Sousedík, Rank-One Inverse Approximations for Hierarchical Stochastic Galerkin Preconditioners Across Polynomial Chaos Models, in preparation.
- **A.K. Aydin**, N. Akhavan, A.S. George, M. Starz-Gaiano, B.E. Peercy, Cell Surface Geometry Guides Three-Dimensional Border Cell Migration in the Drosophila Egg Chamber, in preparation.

RESEARCH EXPERIENCE

Research Assistant | University of Maryland, Baltimore County | Baltimore, MD | Spring 2027
- Supported by NSF DMS #2619580; computational modeling of collective cell migration (supervisor: Dr. Bradford E. Peercy).

Graduate Assistant | University of Maryland, Baltimore County | Baltimore, MD | July 2022 – Present
- Low-rank stochastic Galerkin solvers for unsteady Navier-Stokes flow with uncertain viscosity.
- Rank-one inverse approximations for hierarchical stochastic Galerkin preconditioners.
- Neural reduced-order models for parametrized stochastic PDEs.
- Phase-field models of collective cell migration in 2D and 3D.

Intern | Wolfram | Champaign, IL (Remote) | June 2023 – August 2023
- Built step-by-step Wolfram|Alpha solutions for matrix dimensions, Kronecker products, and LU, Cholesky, and QR decompositions.
- Internship supported through the NSF-funded INMAS program.

Research Assistant | Western Kentucky University | Bowling Green, KY | September 2021 – June 2022
- NSF-supported research on robust boundary sensor design for laminated beam models.
- Supervised undergraduate students developing Wolfram Demonstration Projects.

INVITED TALKS

- **SIAM Conference on Uncertainty Quantification** (UQ26), Minneapolis, Minnesota, March 2026.
- **Joint Mathematics Meetings** (JMM), Washington, DC, January 2026.
- **Mid-Atlantic Numerical Analysis Day**, Philadelphia, Pennsylvania, November 2025.
- **Finite Element Circus**, Arlington, Virginia, October 2025.
- **American Control Conference**, Denver, Colorado, July 2025.
- **AMS Western Sectional Meeting**, San Luis Obispo, California, May 2025.
- **WKU AMS Graduate Student Chapter** talk, Virtual, April 2025.
- **AMS Central Sectional Meeting**, Milwaukee, Wisconsin, April 2024.
- **62nd IEEE Conference on Decision and Control**, Singapore, December 2023.
- **Mid-Atlantic Numerical Analysis Day**, Philadelphia, Pennsylvania, November 2023.
- **SIAM Central States Section**, Lincoln, Nebraska, October 2023.
- **SIAM Conference on Control and Its Applications** (CT23), Philadelphia, Pennsylvania, July 2023.
- **AIMS Conference on Dynamical Systems, Differential Eqs. and Applications**, Wilmington, NC, June 2023.
- **SIAM Southeastern Atlantic Section**, Blacksburg, Virginia, March 2023.
- **Joint Mathematics Meetings** (JMM), Boston, Massachusetts, January 2023.
- **61st IEEE Conference on Decision and Control**, Cancun, Mexico, December 2022.
- **AMS Southeastern Sectional Meeting**, Chattanooga, Tennessee, October 2022.
- **SIAM Pacific Northwest Section**, Vancouver, Washington, May 2022.
- **Joint Mathematics Meetings** (JMM), April 2022.
- **Kentucky Section of the Mathematical Association of America Annual Meeting**, Virtual, April 2022.
- **WKU Student Research Conference**, Bowling Green, Kentucky, March 2022.
- **41st WKU Mathematics Symposium,** Bowling Green, Kentucky, February 2022.
- **Kentucky Academy of Science Annual Meeting**, Virtual, November 2021.

LEADERSHIP

IEEE CSS NextCom | December 2023 – Present
- **General Activities Co-Chair |** 2025 – Present
- **Virtual Ambassador |** 2023 – 2025
- Helped organize the CCTA 2026 NextCom Picnic (Vancouver), the ACC 2025 evening social at Spectra Art Space (Denver), and IEEE CSS Day 2024.
- Involved since NextCom's founding in 2023: "Empowering the Next Generation of CSS Leaders: From NextCom to ExCom."

Graduate Student Association (GSA) | University of Maryland, Baltimore County | July 2024 – October 2026
- **President** | July 2026 – October 2026
- **Vice President** | July 2025 – July 2026
- **Senator** | College of Natural and Mathematical Sciences | July 2024 – July 2025

- Committees: University Steering; Classroom; Graduate Student Organizations; Graduate Assistant Advisory; Janet C. Rutledge Graduate Student Well-Being Award Selection
- UMBC representative to the University System of Maryland Student Council

SIAM Student Chapter | September 2020 – May 2024
- **President** | University of Maryland, Baltimore County | 2022 – 2024
    - Awarded SIAM Student Chapter Certificate of Recognition
- **President** | Western Kentucky University | 2020 – 2022
    - Awarded SIAM Student Chapter Certificate of Recognition

Graduate Council | Western Kentucky University | September 2021 – June 2022

Student Research Grants Committee | Western Kentucky University | September 2021 – June 2022

HONORS & AWARDS

- Outstanding Graduate Teaching Assistant | University of Maryland, Baltimore County | 2026
- SIAM Student Chapter Certificate of Recognition | University of Maryland, Baltimore County | 2024
- John D. Minton Award | Western Kentucky University | 2022
    - The highest university-wide award, given to a single graduate student.
- Outstanding Graduate Student Award, Department of Mathematics | Western Kentucky University | 2022
- Kentucky NSF EPSCoR SuperCollider Best Presentation Award | 2022
- SIAM Student Chapter Certificate of Recognition | Western Kentucky University | 2022
- Dr. James H. Stuteville Scholarship | Western Kentucky University | 2022
- Hugh F. and Katherine A. Johnson Mathematics Scholarship | Western Kentucky University | 2021

RESEARCH FUNDING SUPPORT

- NSF DMS #2619580 | Research Assistantship | University of Maryland, Baltimore County | Spring 2027
- Kentucky NSF EPSCoR, NSF Cooperative Agreement No. 1849213 | Research Assistantship | Western Kentucky University | 2021 – 2022
- NSF-funded INMAS program | Wolfram Internship | 2023

TEACHING EXPERIENCE

Instructor of Record | University of Maryland, Baltimore County | Baltimore, MD | Summer 2024 – Present
- MATH 120 | Introduction to Contemporary Mathematics | Fall 2026 | Online
- MATH 150 | Precalculus | Summer 2026 | Online
- MATH 152 | Calculus II | Spring 2026 | In person
- MATH 221 | Introduction to Linear Algebra | Winter 2026 | Online
- MATH 106 | Algebra and Elementary Functions | Fall 2025 | In person
- MATH 152 | Calculus II | Summer 2025 | In person
- MATH 151 | Calculus I | Summer 2024 | In person

Graduate Assistant | University of Maryland, Baltimore County | Baltimore, MD | July 2022 – Present
- Teaching assistant in semesters without an instructor-of-record assignment (e.g., MATH 152, Fall 2024).
- Completed CIRTL (Center for the Integration of Research, Teaching, and Learning) certification at the Associate and Practitioner levels.

Orientation Advisor | University of Maryland, Baltimore County | Baltimore, MD | March 2024 – Present
- Advised more than 300 incoming undergraduate students on prior coursework, majors, course selection, and schedules supporting progress toward graduation.

Teaching Assistant | Western Kentucky University | Bowling Green, KY | September 2020 – July 2021
- Tutored at the Mathematics Testing Center in Summer 2021.
- Completed Graduate Teaching Assistant Institute (GATI) modules I & II.
- Awarded a Summer 2021 assistantship; received an offer to serve as a Graduate Instructor.

Student Teacher | Haydar Akın Vocational High School | Istanbul, Turkey | December 2018 – June 2019
- Taught 9th- and 10th-grade mathematics for a semester under the supervision of a mathematics teacher and a pedagogical expert, as part of the Pedagogical Formation program.

WORKSHOPS

- CIRTL Teaching-as-Research, Summer 2025
  Proposal: Developing Students' Symbol Sense with Targeted Practice
- 62nd IEEE Conference on Decision and Control, Singapore, December 2023.
  Modern Adaptive Control and Estimation: From Theory to Applications.
- SIAM Mathematical Problems in Industry Workshop (MPI 39), NJIT, New Jersey, June 2023.
  Report: Bubbles in Flow Streams and Porous Media (industrial partner: W. L. Gore & Associates).
- 18th Annual Graduate Student Mathematical Modeling Camp, University of Delaware, June 2023.
  Report: Modeling of Drying Process in Porous Media.

WOLFRAM DEMONSTRATION PROJECTS

Wolfram Demonstration Projects are reviewed, interactive Wolfram notebooks that explore mathematical concepts.

- J. Walterman, **A.K. Aydin**, A.Ö. Özer, Stabilization of a Smart Beam with a Tip Mass, 2024.
- J. Walterman, **A.K. Aydin**, A.Ö. Özer, Rapid Stabilization of Heat and Structure Interactions with Boundary Feedback Controllers, 2024.
- J. Walterman, **A.K. Aydin**, S. Leveridge, A.Ö. Özer, Dynamics of a Longitudinal Piezoelectric Beam, 2023.
- J. Walterman, **A.K. Aydin**, M. Poynter, A.Ö. Özer, Stabilization of the Wave Equation by Direct Fourier Filtering, 2022.
- **A.K. Aydin**, M. Poynter, A.Ö. Özer, Vibration Suppression on a Hinged Three-Layer Sandwich Beam, 2022.
- **A.K. Aydin**, M. Poynter, A.Ö. Özer, Feedback Sensor Design for a Cantilevered Three-Layer Sandwich Beam, 2022.
- M. Poynter, L. Stewart, **A.K. Aydin**, A.Ö. Özer, Boundary-Feedback Control of Vibrations on a String with and without Filtering, 2022.

SERVICE

- Nonlinear Analysis: Real World Applications | 2026 – Present | Reviewer
- American Control Conference (ACC) | 2025 – Present | Reviewer
- AMS Mathematical Reviews / MathSciNet | 2025 – Present | Reviewer
- IEEE Control Systems Letters | 2024 – Present | Reviewer
- IEEE Conference on Decision and Control (CDC) | 2024 – Present | Reviewer
- Professional memberships: AMS, SIAM, IEEE

SKILLS

- **Programming:** MATLAB, Python, Mathematica/Wolfram Language
- **Numerical methods:** finite elements, finite differences, stochastic Galerkin and polynomial chaos, Krylov solvers and preconditioning, tensor-train and CP decompositions, reduced-order modeling
- **Analysis and control:** boundary observability, Lyapunov stability, feedback and sensor design
- **Languages:** English, Turkish

REFERENCES

- **Bedřich Sousedík**, University of Maryland, Baltimore County (Ph.D. advisor) | sousedik@umbc.edu
- **Ahmet Özkan Özer**, Western Kentucky University (M.Sc. advisor) | ozkan.ozer@wku.edu
- **Andrei Draganescu**, University of Maryland, Baltimore County | draga@umbc.edu
- **Justin Thomas Webster**, University of Maryland, Baltimore County | websterj@umbc.edu